

BOLT ABACUS

Product Requirements Document

Student Learning Portal · v1.0

Document Control

Field	Detail
Status	DRAFT
Version	1.0
Date	May 2026
Author	Product Team
Reviewers	Design, Engineering, QA
Figma Reference	Bolt Abacus V1 Design

1. Executive Summary

Bolt Abacus is a web-based student learning portal targeted at children in Grades 1–4 who are enrolled in structured abacus mathematics programs. The product must deliver two core experiences — structured **Learn** journeys and skill-building **Practice** sessions — inside a single, cohesive platform.

The design direction, confirmed in the Figma file **Bolt Abacus V1 Design**, positions the product at the intersection of a gaming platform and an edtech tool: dark charcoal backgrounds, bright yellow (#F2C94C) primary accents, glassmorphism card surfaces, and a vocabulary borrowed from competitive gaming ("Training Arena", "Path of Conquest", "Mission Passed"). This language is intentional — it makes the product feel aspirational and exciting for the target age group without descending into generic children's-app cartoon aesthetics.

1.1 Strategic Goals

- **Engagement:** Make every session feel like a game, not a worksheet.
- **Progression:** Give students a clear sense of advancement through visual rewards and unlockable content.
- **Simplicity:** Max 2–3 clicks from login to active learning.
- **Trust:** Parents and teachers should immediately understand the structure and rigor of the curriculum.

1.2 Figma Design Validation

The Figma file ([wMgk2iCT5zY3uPd1wCI9Ju](#)) contains 8 fully designed screens across one page, confirming the PM handoff is visually resolved at high fidelity for all major flows. No screens are missing from the handoff document.

Figma Frame	Description	Status
Bolt Abacus Hub	Landing / Home screen	✓ Complete
Level Selection	Learn – grid view of 10 levels	✓ Complete
The Training Arena	Practice – mode selection	✓ Complete
Path of Conquest	Learn – vertical Duolingo-style path	✓ Complete
Arena Setup	Practice – configuration screen	✓ Complete

In the Arena	Active practice session	✓ Complete
Classwork Practice	Learn – classwork session screen	✓ Complete
Classwork Report	Post-session results + breakdown	✓ Complete
Victory!	Practice results screen	✓ Complete

1.3 Gaps & Discrepancies (PM Doc vs. Figma)

The following items appear in the PM handoff but are NOT yet in the Figma file:

- **Login / Auth Screen** — PM handoff references a login entry point but no login screen exists in the Figma.
- **Advanced Levels (11–12)** — PM spec calls for "+2 advanced levels" beyond Level 10; Figma level grid only shows Levels 1–10.
- **Victory screen for Practice (free mode)** — The "Victory!" frame exists but flows from Classwork. A separate Practice victory state is not defined.
- **Flash Cards mode UI** — "Memory Master" card exists in the Training Arena but its dedicated session screen is not designed.
- **Zen Mode / No Rush Mastery session screen** — Similarly, no dedicated session UI for this mode.

These items are flagged as **P1 design debt** and must be resolved before engineering begins those flows.

2. Product Overview

2.1 Target Audience

Attribute	Detail
Primary Users	Students, ages 5–10, Grades 1–4
Secondary Users	Parents (oversight) and Abacus instructors (content assignment)
Literacy Level	Low – UI must communicate through icons, color, and short single-word labels wherever possible
Device	Desktop / large-screen browser (primary); tablet support desirable in v2
Session Length	Expected 10–30 min per session; UI must support graceful interruption and resume

2.2 Design System – Tokens Confirmed in Figma

Token	Value	Usage
Background	#1A1A1A / #0F0F0F	Page background; dark charcoal
Surface	#262626 – #2D2D2D	Card and panel backgrounds
Primary Accent	#F2C94C (Yellow)	CTA buttons, active states, headings, icon fills
Electric Blue	#0EA5E9	Zen Mode / "No Rush" card accent
Purple	#8B5CF6	"The Lab" / Custom Challenge card accent
Neon Orange	#F97316	"Sonic Speed" / Time Attack card accent
Success	#22C55E	Correct answer feedback; completion checkmarks
Error	#EF4444	Wrong answer feedback; locked state text
Text Primary	FFFFFF	Main content text on dark surfaces
Text Secondary	#9CA3AF	Subtitles, helper text, locked state labels
Border	#3F3F46	Card borders; dividers
Glassmorphism	rgba(255,255,255,0.06)	Active card overlays; problem container backdrop

2.3 Typography

- **Display / Hero:** Large all-caps condensed font (as seen in "WELCOME BACK, CHAMPION!" and "PATH OF CONQUEST"). Figma renders these at ~64–72px with letter-spacing expanded. Recommendation: **Bebas Neue** or similar display font.
- **Headings (H2–H3):** Bold sans-serif, white or yellow. Figma uses sizes 28–36px for section titles.
- **Body / Labels:** Regular sans-serif. Figma renders all body copy and UI labels in a neutral sans (consistent with Inter or similar system font).
- **Monospace / Numeric:** The math problem display ("1,452 – 834 + 210") uses large tabular numerics. Ensure the chosen font has tabular number support for vertical alignment.

3. Information Architecture

The application has a deliberately flat navigation structure. After login, all flows branch from a single Hub screen. There is no persistent sidebar or bottom nav — each section owns its own contextual top bar.

Hub → **[LEARN]** → Level Selection (Grid) → Path of Conquest (Vertical) → Classwork Session → Mission Report

Hub → **[PRACTICE]** → Training Arena → (Mode card) → Arena Setup (Custom only) → In the Arena → Victory Screen

3.1 Navigation Components Confirmed in Figma

Component	Location	Elements
Top App Bar (Hub)	Hub screen	Brand logo ("BOLT ABACUS" in yellow); XP stats icons (⚡ 🔥 🏆); Level badge ("LEVEL 12"); Avatar
Top App Bar (Learn)	Level Selection, Path	← Back arrow; Brand logo; Same stats strip + Level badge + Avatar
Top App Bar (Practice)	Training Arena	Brand logo; larger stats strip; avatar with border glow
Task Bar (In the Arena)	Active session	"← ABANDON RUN" link (left); Timer pill (center-right); Question counter pill (far-right). Full-width, nav stripped.
Breadcrumb	Classwork session	Small pill showing "Level X > Classwork"; progress label below

4. Hub Screen (Bolt Abacus Hub)

Figma Frame ID: 1:2. The entry point after authentication. Must feel like a "mission control" room — immersive, high energy, but never cluttered.

4.1 Layout Specification

- **Hero Greeting:** Subtext "TRANSMISSION RECEIVED" (spaced caps, muted yellow) above a massive hero heading "WELCOME BACK, CHAMPION!" in display font. The exclamation mark is part of the designed text — preserve casing and punctuation exactly.
- **Accent Rule:** A short horizontal yellow gradient bar sits below the hero text as a visual separator (visible in Figma at ~96px wide, centered).
- **Portal Cards:** Two full-height cards side by side (each ~516px wide × 512px tall at 1280px viewport). Equal width split with a 40px gap.
- **Ambient Particles:** Three CSS-simulated dot elements (8px, 12px, 6px) at fixed positions to add depth. No JS animation required for v1.

4.2 Portal Card Specification

Property	LEARN Card	PRACTICE Card
Icon	Compass / Navigation icon (30×30)	Gamepad icon (30×21)
Icon Container	64×64 pill, yellow-tinted border	64×64 pill, blue-tinted border
Heading	"LEARN" – display font, white	"PRACTICE" – display font, white
Subtext	"Master the Abacus"	"Level up your speed"
Background	Gradient: dark → yellow-amber glow upper-right	Gradient: dark → blue-grey glow upper-right
Overlay	Glassmorphism blur circle (upper-right)	Same pattern, lighter tint
Hover State	Border brightens; subtle scale transform (1.02)	Same
Navigation	→ Level Selection screen	→ Training Arena screen

4.3 Acceptance Criteria

- Both cards are full-height and equal-width at 1280px viewport.
- Cards are keyboard-navigable (Tab / Enter / Space).
- Hero text "WELCOME BACK, CHAMPION!" renders in display font at ≥60px.

- Ambient particles are present and non-interactive.
- Clicking either card navigates within $\leq 300\text{ms}$ (no full page reload if SPA).

5. Learn Flow

5.1 Level Selection Screen

Figma Frame ID: 1:58. This is a grid-based "overview" of all 10 levels (+ 2 advanced). Students see where they are and where they are going at a glance.

5.1.1 Page Header

- "PATH OF CONQUEST" — display font, yellow, centered.
- Sub-heading: "Master the Abacus, Conquer the Realms." — regular weight, muted white.

5.1.2 Level Grid Layout (Figma-Confirmed)

A 5-column grid renders Levels 1–10 (plus 2 advanced). Each cell is ~211px wide × 204px tall with 24px gutters. Two rows of five. The Current Mission card (Level 4 in Figma) is ~221×214px — slightly larger — and renders with a bright yellow border glow and a "CURRENT MISSION" badge.

State	Visual Treatment	Icon	CTA
Completed	Yellow gradient border; full opacity	✓ Checkmark (gold)	Level # + Topic name only (no button)
Current Mission	Yellow border glow; slightly larger card; "CURRENT MISSION" pill badge	▶ Play icon (in circular button)	"ENGAGE" yellow CTA button
Locked	Dimmed surface; muted text; no gradient	🔒 Lock icon (grey)	None — card is not clickable

5.1.3 Level Names (Confirmed in Figma)

#	Topic Name	#	Topic Name	State
01	Foundations	06	Double Digits	Locked
02	The Big Friend	07	Mind Meld	Locked
03	Speed Striker	08	Mental Matrix	Locked
04	The Little Friend	09	Zenith Trial	Locked
05	Combo Strike	10	Grandmaster	Locked

Note: Levels 11–12 ("Advanced") are referenced in PM doc but not present in Figma.

Names TBD by curriculum team.

5.2 Path of Conquest Screen (Vertical View)

Figma Frame ID: 1:272. Accessed when a student "engages" a level. A Duolingo-style vertical progression path showing individual lesson nodes for that level.

5.2.1 Layout

- Centered 672px column on a 1280px canvas.
- Each "Class" (grouping of lessons) is introduced by a horizontal divider pill showing "CLASS 1 — 3/4" (class name and completion fraction).
- Lesson nodes are circular buttons (64×64) connected by a 4px wide vertical line. The line is yellow for completed segments and dark/muted for locked segments.
- Completed nodes: yellow circle + checkmark icon.
- Current/Active node: larger (80×80) with a play icon; expands into a detailed card below it.
- Locked nodes: semi-transparent circle + lock icon.

5.2.2 Active Level Card (Expanded)

When the student's current level is active, the node expands into a 512×343px card containing:

- "CURRENT MISSION" pill badge (top).
- Level number (large display) + Topic name below.
- **Start Classwork** — Primary yellow CTA button (full width).
- **Homework [Locked]** — Secondary ghost button (disabled until Classwork complete). Figma shows lock icon left of label.
- **Download Materials** — Tertiary text link with download icon.

5.2.3 Progression Logic

Trigger	Effect
Classwork completed	Homework button unlocks on the same level card
Homework completed	Next level node unlocks; path connector turns yellow
Level 10 Homework done	Advanced Levels 11–12 unlock (details TBD)
"Continue where left off"	App scrolls path to the active node on load

5.3 Classwork Session Screen

Figma Frame ID: 1:619. The active practice environment inside a Learn level.

5.3.1 HUD Row (Header)

Left	Center	Right
Breadcrumb pill ("Level X > Classwork") + large heading "Classwork" + progress indicator	Question progress bar with fraction (e.g. "Q 5 / 30") centered above the bar	Timer HUD — clock icon + MM:SS display

5.3.2 Problem Canvas

- Glassmorphic card (768×275px) centered on the canvas.
- Math problem displayed vertically: operands stacked right-aligned, operators (−, +) to the left in muted color.
- Horizontal divider line before the answer row.
- Input field (160×100px) — large, with keyboard icon and "?" placeholder.

5.3.3 Action Controls

- **Submit** — Primary yellow full-width button below the card.
- **Skip** — Ghost text link (left of controls).
- **Hint** — Small ghost pill (right of controls). Figma shows a key icon + "Hint" label.

5.3.4 Answer Feedback

State	Visual Treatment	Label
Correct	Green pill with check icon; brief glow on card border	"+150 XP" (Figma confirmed)
Wrong	Red/orange pill with warning icon; shake animation on card	"RECALCULATE" (Figma confirmed)

6. Practice Flow

6.1 Training Arena Screen

Figma Frame ID: 1:183. Mode selection hub for all practice types. Branded as "TRAINING ARENA". Sub-heading: "Select your protocol. Master the Abacus."

6.1.1 Mode Cards (Bento Grid)

Four cards in a 2×2 bento grid (each 564×275px). Each card has a unique color accent and icon.

Mode	Label (Figma)	Accent Color	Description (Figma copy)
Flash Cards	FLASH CARDS / "Memory Master"	Yellow (#F2C94C)	"High-velocity visual recognition training to build unshakeable abacus memory."
No Rush Mastery	NO RUSH MASTERY / "Zen Mode"	Blue (#0EA5E9)	"Perfect your technique without the pressure of time. Focus on absolute precision."
Time Attack	TIME ATTACK / "Sonic Speed"	Orange (#F97316)	"Race against the clock in high-stakes computational challenges."
Custom Challenge	CUSTOM CHALLENGE / "The Lab"	Purple (#8B5CF6)	"Configure parameters, set digit limits, and create your ultimate training ground."

Card layout pattern (Figma-confirmed):

- Icon container (64×64) — rounded square, accent-colored background with subtle border + shadow.
- Mode label in spaced uppercase caps (e.g. "FLASH CARDS") — accent color.
- Marketing title (e.g. "Memory Master") — large, white, 37px.
- Description paragraph — muted white, 2 lines max.
- Small arrow icon (top-right corner of card) — indicates clickability.

6.2 Arena Setup Screen (Custom Challenge)

Figma Frame ID: 1:397. Shown when the student taps "The Lab". Other modes (Flash Cards, Zen Mode, Sonic Speed) use preset configs and skip this screen.

6.2.1 Operations Selector

- 4 large button tiles in a horizontal row: Addition (+), Subtraction (−), Multiplication (×), Division (÷).
- Selected state: yellow background + yellow border (Addition is selected in Figma).
- Unselected: dark surface + dark border.
- Single-select (Figma v1) — only one operation active at a time.

6.2.2 Numerical Parameters (Left Column)

Parameter	Min	Max	Default (Figma)
Time Limit (min)	1	60	1 minute
Questions	5	50	10
Digits	1	5	1
Rows	2	10	2

Each parameter uses a card-style slider with icon + label + current value badge (right). Range labels below the slider track.

6.2.3 Advanced Rules (Right Column — Figma-confirmed toggles)

Toggle	Default State & Behavior
Include Subtraction	ON (yellow) — mix subtraction into addition sessions
Zig-Zag Pattern	OFF — alternates between larger and smaller numbers
Same Digits in Answer	OFF — constrains generated answers to same digit count
Audio Mode	OFF — questions are read aloud; student types answer

6.2.4 CTA

- **"ENTER ARENA"** — Full-width primary yellow button at bottom. Rocket icon left of text. Font size ~32px. Navigates to In the Arena.

6.3 In the Arena (Practice Session Screen)

Figma Frame ID: 1:553. The focused gameplay experience. Navigation is deliberately suppressed — only "← ABANDON RUN" is available.

6.3.1 Task Bar (Header)

- **Left:** "← ABANDON RUN" — small text link (exit route back to Training Arena).

- **Right:** Timer pill ("00:42" with clock icon) + Question counter pill ("Q 5 / 10" with target icon).

6.3.2 Progress XP Bar

- A 448px horizontal bar centered on the canvas showing session progress. Yellow fill on dark track. Positioned above the problem card.

6.3.3 Problem Card

- 448×430px glassmorphic card, centered.
- Math operands stacked vertically, right-aligned. Operators (−, +) left-aligned in muted color. Large numeric font (~52px).
- Horizontal rule divides problem from answer zone.
- Answer input: wide field, keyboard icon left, "?" placeholder right.

6.3.4 Submit CTA

- **"ENGAGE ►"** — Yellow pill button below the card (~229px wide × 52px).

6.3.5 Feedback Indicators (Figma confirmed, shown after submission)

- **+150 XP** — Green pill (correct).
- **RECALCULATE** — Orange-red pill with warning icon (wrong).

7. Results Screens

7.1 Classwork / Mission Report

Figma Frame ID: **1:678**. Shown immediately after completing a Classwork session.

7.1.1 Status Header

- **"MISSION PASSED"** — Display font, yellow, centered. Sub-label: "Level 12 Cleared" (or relevant level).

7.1.2 Stats Grid (3 Cards)

Score Card	Accuracy Card	Time Card
Icon: Medal/trophy	Icon: Target (concentric rings)	Icon: Stopwatch
Label: "SCORE"	Label: "ACCURACY"	Label: "TIME TAKEN"
Value: "28/30" (example)	Value: "93%"	Value: "04:45"

7.1.3 XP Progress Bar

- "XP GAINED" label left; "+450 XP" right (yellow, prominent).
- Filled bar showing session progress toward next level.
- "1,200 XP to Level 13" — helper text right-aligned below bar.

7.1.4 Question Breakdown Table

VERDICT	QUESTION	YOUR ANSWER
Correct ✓	33 + 11 - 11	33
Wrong ✗	88 - 33 + 22	72
Wrong ✗	66 + 22 - 55	324324

7.1.5 Action Buttons

- **"NEXT MISSION"** — Primary yellow, ~286px wide. Navigates to next level/classwork.
- **"RETAKE SESSION"** — Ghost/outline button, ~290px wide. Restarts the same session.

7.2 Victory Screen (Practice)

Figma Frame ID: **1:795**. Shown after completing a free-practice session.

7.2.1 Header

- **"Victory!"** — Display font. Sub-label: "Level cleared" or session summary.

7.2.2 Stats Bento Grid

Accuracy Card	Main Score Card	Speed Card
Decorative icon top-right (out of card bounds)	Highlighted (gradient border); center of grid	Decorative icon bleeding left edge
Big % value + "ACCURACY" label	Large score value + label	Speed value (avg sec/question) + "SPEED" label

7.2.3 XP Progress

- Same pattern as Mission Report: label + bar + next-level XP threshold.

7.2.4 Action Buttons

- **"← Play Again"** — Returns to Training Arena mode card (same mode).
- **"Back ►"** — Returns to Training Arena hub (mode selection).

8. Gamification & XP System

The Figma designs surface several gamification elements that require backend and frontend engineering effort. This section formalizes the requirements inferred from the design.

8.1 XP Points

- XP is earned on completion of Classwork sessions, Homework sessions, and Practice sessions.
- XP earned per session is shown in the results screen ("+450 XP").
- A running XP total drives a player Level (confirmed: "LEVEL 12" shown in Figma nav bar).
- The XP bar on results screens shows progress to next player Level (e.g. "1,200 XP to Level 13").

Open Question: What is the XP required per level? XP bonuses for streaks or accuracy? These must be defined by the curriculum/product team before engineering begins.

8.2 HUD Stats Icons

The top navigation bar shows three icons (⚡ 🔥 🏆) representing real-time stats. Their exact meaning is not specified in the PM handoff. Based on gaming conventions and Figma layout:

- ⚡ **(Lightning bolt):** Current XP or Energy points.
- 🔥 **(Flame):** Streak counter (consecutive correct answers or daily login streak).
- 🏆 **(Trophy):** Achievements unlocked or Rank.

Action Required: Product/Curriculum team to confirm the mapping of these three icons before sprint planning.

8.3 Level Completion States

- Node changes from "active" (play icon) to "completed" (gold checkmark) upon Homework completion.
- Vertical path line segment turns yellow upon completion.
- These states must persist across sessions (backend-driven progress API).

9. Technical Requirements

9.1 Performance

- Time to Interactive (TTI) $\leq$ 3 seconds on 4G mobile connection.
- Practice session page transitions $\leq$ 300ms.
- Animation and hover effects must not drop below 60 fps on target hardware.
- Glassmorphism / backdrop-filter effects must degrade gracefully on browsers without support (fallback: semi-transparent solid background).

9.2 Accessibility

- WCAG 2.1 AA compliance minimum (though target users are children, accessibility benefits all).
- All interactive elements keyboard-navigable.
- Color is never the sole differentiator for state (locked/unlocked icons must differ in shape as well).
- Timer HUD must have a screen-reader-friendly live region.
- Math problems must be readable by screen readers (not rendered as images).

9.3 Browser & Device Support

Priority	Browser / Device	Target
P1 (must)	Chrome 120+ (desktop)	1280px+ viewport
P1 (must)	Firefox 120+ (desktop)	1280px+ viewport
P2 (should)	Safari 17+ (desktop/iPad)	1024px+ viewport
P2 (should)	Chrome (Android tablet)	1024px+ viewport
P3 (nice-to-have)	Mobile portrait	v2 scope

9.4 PDF Download

- Each level in the Path of Conquest has a "Download Materials" link.
- PDFs are pre-generated question sheets, not dynamically rendered.
- Files should be named: bolt-abacus-level-{N}-{classwork|homework}.pdf.
- Download triggers the browser native download flow (no custom UI required).

10. Open Items & Risks

#	Item	Description	Owner	Priority
1	Login / Auth Screen	No design exists. Needed before any other screen is reachable.	Design	P0 – Blocker
2	Advanced Levels 11–12	Referenced in PM doc; not in Figma. Names and content TBD.	Curriculum + Design	P1
3	Flash Cards session screen	Mode card designed; session UI missing.	Design	P1
4	Zen Mode session screen	Mode card designed; untimed session UI missing.	Design	P1
5	HUD icon mapping	⚡ 🔥 🏆 meaning undefined. Needed for API and display logic.	Product	P1
6	XP formula	XP per session, per level, bonuses not defined.	Product	P1
7	Practice Victory (free mode)	Separate victory state for free practice vs Learn not confirmed.	Design	P2
8	Audio Mode implementation	Toggle exists in Setup; TTS engine and question audio format TBD.	Engineering	P2
9	Mobile responsiveness	All Figma frames at 1280px. No mobile breakpoints designed.	Design	P3 – v2
10	Multi-operation selection	Setup screen shows single-select. PM doc may intend multi-select.	Product	P2

11. Success Metrics

Metric	Target	Measurement Method
Session start rate (Hub → first action)	≥ 90%	Analytics: Hub load vs card click
Time to first practice submission	≤ 90 sec	Timer from Hub load to first "ENGAGE" click
Homework unlock rate (% who complete Classwork)	≥ 70%	Progress API events
Practice session completion rate	≥ 65%	Sessions started vs Mission Report loaded
Weekly active users (WAU)	Baseline TBD post-launch	Auth log events
NPS (students 7–10 y/o adapted survey)	≥ 50	In-app prompt post session #5

Appendix A – Figma Frame Reference

Frame ID	Frame Name	Canvas Position	Notes
1:2	Bolt Abacus Hub	0, 0	Landing screen; 1280×1025
1:58	Level Selection	0, 1191	Learn grid; 1280×1024
1:183	The Training Arena	1639, 32	Practice hub; 1280×1024
1:272	Path of Conquest	0, 2381	Vertical lesson path; 1280×2061 (tall)
1:397	Arena Setup	3187, 32	Custom practice config; 1280×1293
1:553	In the Arena	4640, 0	Active session; 1280×1024
1:619	Classwork Practice	0, 4688	Learn session; 1280×1024
1:678	Classwork Report	0, 5958	Mission Report / Results; 1280×1211
1:795	Victory!	6093, 0	Practice results; 1280×1024

Appendix B – Revision History

Version	Date	Author	Changes
1.0	May 2026	Product Team	Initial PRD — converted from PM handoff + Figma cross-check